

Kartik Ramachandran

[Portfolio](#) | [LinkedIn](#) | kramacha@uwaterloo.ca | (825)-735-8445

Skills

CAD - Solidworks, Autodesk Inventor, Fusion 360, AutoCAD
Languages - C++, Python, Matlab, VEX PROS
Manufacturing - CNC Cutting, 3D Printing, Welding, Mill, Lathe, Various Power Tools

Education - University of Waterloo

Systems Design Engineering

Expected Completion 2030

Projects:

Exia Labs: Argus Automated Military Vehicle

December 2025 – January 2026

- Engineered a modular **Unmanned Ground Vehicle (UGV)** to compete in the NATO xTech Military Competition in Germany, designed to **autonomously carry 200+ kg payloads** across all terrain types using autonomous navigation using ROS-based path planning and LiDAR object detection, implemented in C++.
- Reverse engineered a 2007 Suzuki KingQuad ATV, isolating and repurposing the drivetrain, transmission, throttle and braking systems for **full autonomous operation**. Replaced steering with **ODrive autonomous turning** through **CNC'd steel gears**.
- Conducted **shear force and moment calculations** to determine steel chassis support locations, **welded and angle-ground** a frame validated to 200 kg, **automated gas throttle** with **120 kg*cm servos**, and fabricated a ventilated steel shell to protect onboard electronics in harsh outdoor environments

VEX Robotics Team 210Z

August 2021 – May 2024

- Led end-to-end design and integration of **25+ competition robots**, 6 motor drivetrains, developing **PTO transmissions**, **4-bar lifts**, **silicon wheel systems**, flex-wheel intakes, and pneumatic endgame mechanisms for climbing and expanding.
- Developed and tuned closed-loop motion control, **Pure Pursuit path tracking**, and **odometry** to improve autonomous pathing.
- Ranked **first nationally**, seventh in autonomous skills at the **US Open** and **finished within the top 0.26% internationally**.

Experience:

PowerHouse Data Centre Group - Data Centre Thermal Design Intern

January 2026 – May 2026

- Utilized **AutoCAD** to develop site expansion drawings for a **TELUS data centre**, designing thermal and computing infrastructure, and produced **Electrical drawings** for Uninterruptible Power Supply (UPS) systems, rated **300 VA to 5,000**.
- Engineered and specified **Vertiv/Liebert thermal cooling** solutions for clients across a wide cooling load range (**5kW– 265kW**), supporting environments where rack densities routinely **exceed 100 kW** in high-performance facilities.
- Created engraved lamacoid identification from electrical drawings for **power cabinets at a Satellite facility** in High River, AB - ensuring compliance with **Canadian Electrical Code (CEC)** requirements for clear identification in emergencies.

Western Mechatronics Robotics

Robotics Executive

May 2021 – June 2024

- Co-Leader of a globally recognized robotics education company with **\$240,000 ARR**, **156 members** and a **7,000+ sqft workshop** dedicated to creating **robotics and leadership** opportunities for **under-represented youth**.
- Secured partnerships with **Google, TC Energy, Telus and Garmin** to host events for underrepresented groups in robotics, leading to a **260% increase in Alberta Robotics Participation** since 2019, and the creation of VEX University in Alberta.

Events/Outreach/Inclusion Planning

May 2022 – Current

- Organized 8 annual outreach programs with **1000+ attendees**, hosting workshops at **The Telus Spark Sciences Center, GMC Stadium and the University of Calgary**. Led **Calgary Stampede's** first robotics booth and first robotics tournament, inviting best teams internationally, promoting STEM to **1,500,000+ visitors** in two weeks, participants **building 3100+ robots**.

NEXT Robotics

July 2022 – April 2024

- Designed and built multiple **STEM based curriculums** taught to students ranging from **Grades 2 – 12**, focusing on engineering principles and coding basics. Facilitated **full-time summer camps** and year-round classes, allowing students to **explore robotics** and prepare for competitive team, **impacting 350+ students**.

Annual Girl Powered Robotics Workshop

May 2023 – Present

- **Co-led** Canada's first Girl Powered Robotics Workshop alongside **Google and University of Calgary**, providing girls in middle and high school the opportunity to **engage with STEM** regardless of their circumstances
- **Created curriculums**, and contacted **robotics experienced presenters** with experiences and inspirations to share. Designed **custom blueprints** for 250+ students, helping **construct, code** and **operate** the robots to participate in friendly competitions

Mecha Mayhem International Robotics

August 2024 – Present

- Co-organized **Canada's largest robotics competition**, bringing together the 300+ best teams (1,500+ competitors) internationally. **Managed logistics** across multiple divisions running simultaneously and helped **fundraise \$200,000+**.
- **Led 250+ volunteers** onsite in overseeing the operation of the event, working with sponsors and event donors.

UWaterloo Formula-Electric: Chassis Team Member

September 2025 – Present

- 3D-modelled and printed **Joint Tolerance Measuring** tool to determine the angle at steel joints on the **main chassis**
- Assisted with chassis development and completed workshop power-tool safety courses